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INVITED REVIEW



Technology-Enabled Surveillance as a Regime Characteristic: Re-Authoritarianization in Central Asia

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Abstract

Rapidly advancing information and communication technologies, along with the proliferation of artificial intelligence and closed-circuit television, have profoundly reshaped state-society relations worldwide. Initially celebrated for their democratizing potential, these technologies have, over the past two decades, ushered in a wave of technology-enabled re-authoritarianization, with even established democracies now “at risk.” While numerous terms—such as digital authoritarianism and digital repression—seek to capture this shift, political science has largely neglected the structural role of technology-enabled surveillance. To address this gap, we propose extending Juan Linz and Alfred Stepan's (1996) seminal typology of regime characteristics by adding technology-enabled surveillance as a fifth dimension. To illustrate the empirical relevance of this conceptual extension, the contributions to our special section focus on Central Asia—a region marked by both rapid digitalization and authoritarian resurgence. They show how technological infrastructure is no longer a neutral tool but an essential regime characteristic.

KEY WORDS

Authoritarianism, Central Asia, Democratic backsliding, Linz & Stepan, Surveillance, Technology

Zusammenfassung

Die rasante Entwicklung von Informations- und Kommunikationstechnologien sowie die Verbreitung von Künstlicher Intelligenz und Videoüberwachung haben

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das Verhältnis zwischen Staat und Gesellschaft weltweit tiefgreifend verändert. Zwar wurden diese Technologien zunächst für ihr demokratisierendes Potenzial gelobt, doch in den vergangenen zwei Jahrzehnten haben sie eine Welle der Autokratisierung hervorgerufen, die selbst etablierte Demokratien bedroht. Zahlreiche Begriffe – wie etwa „digitaler Autoritarismus“ oder „digitale Repression“ – versuchen, diesen Wandel zu erfassen, doch hat die Politikwissenschaft die strukturelle Bedeutung technologiegestützter Überwachung bislang vernachlässigt. Um diese Lücke zu schließen, erweitern wir die einflussreiche Regimetypologie von Juan Linz und Alfred Stepan aus dem Jahr 1996 um eine fünfte Dimension: technologiegestützte Überwachung. Zur empirischen Untermauerung dieser konzeptionellen Erweiterung konzentrieren sich die Beiträge unserer „Special Section“ auf Zentralasien, eine Region, die sowohl durch rasante Digitalisierung als auch Autokratisierung geprägt ist. Unsere AutorInnen zeigen auf, dass technologische Infrastrukturen kein neutrales Werkzeug mehr darstellen, sondern zu einem zentralen Regimemerkmal geworden sind.

Résumé

Le progrès rapide des technologies de l'information et de la communication, ainsi que la prolifération de l'intelligence artificielle et de la surveillance vidéo, ont profondément remodelé les relations entre les États et les sociétés dans le monde entier. D'abord célébrées pour leur potentiel de démocratisation, ces technologies ont, au cours des deux dernières décennies, donné naissance à une vague de réautoritarisation par la technologie, même les démocraties établies étant désormais « en danger ». Alors que de nombreux termes, tels que l'autoritarisme numérique et la répression numérique, tentent de rendre compte de ce changement, la science politique a largement négligé le rôle structurel de la surveillance facilitée par la technologie. Pour combler cette lacune, nous proposons d'étendre la typologie des caractéristiques des régimes de Juan Linz et Alfred Stepan (1996) en ajoutant la surveillance facilitée par la technologie comme cinquième dimension. Pour illustrer la pertinence empirique de cette extension conceptuelle, les contributions à notre section spéciale se concentrent sur l'Asie centrale, une région marquée à la fois par une numérisation rapide et une résurgence de l'autoritarisme. Elles montrent que l'infrastructure technologique n'est plus un outil neutre mais une caractéristique essentielle du régime.

Riassunto

Il rapido progresso delle tecnologie dell'informazione e della comunicazione, insieme alla proliferazione

dell'intelligenza artificiale e della videosorveglianza a circuito chiuso, ha profondamente rimodellato le relazioni tra Stato e società a livello globale. Inizialmente celebrate per il loro potenziale democratizzante, negli ultimi due decenni, queste tecnologie hanno alimentato a un'ondata di ri-autoritarizzazione abilitata dalla tecnologia, mettendo “a rischio” persino democrazie consolidate. Sebbene numerosi concetti - come autoritarismo digitale e repressione digitale - tentino di cogliere questo cambiamento, la scienza politica ha in larga misura trascurato il ruolo strutturale svolto dalla sorveglianza tecnologica. Per colmare questa lacuna, proponiamo di estendere la fondamentale tipologia di caratteristiche dei regimi elaborata da Juan Linz e Alfred Stepan (1996), aggiungendo la sorveglianza abilitata dalla tecnologia come quinta dimensione. Per illustrare la rilevanza empirica di questa estensione concettuale, i contributi della nostra sezione speciale si concentrano sull'Asia centrale, una regione segnata sia da una rapida digitalizzazione che da una rinascita autoritaria. Essi mostrano come l'infrastruttura tecnologica non sia più uno strumento neutro, ma una caratteristica essenziale del regime.

INTRODUCTION

Rapidly advancing information and communication technologies (ICTs), along with the proliferation of artificial intelligence (AI) and closed-circuit television (CCTV), have profoundly reshaped state-society relations and governance around the world. Initially celebrated for their democratizing potential—enabling new forms of civic engagement, protest mobilization, and political participation (Earl & Kimport, 2011; Carr & Hayes, 2015; Roberts & Oosterom, 2024)—these technologies are now equally associated with disinformation, mass surveillance, and digital repression (Dall'Agnola, 2024; Roberts & Oosterom, 2024). Governments and corporations alike deploy digital tools to monitor behaviour, manipulate opinion, and curtail dissent (Glasius & Michaelsen, 2018; Zuboff, 2019, 2022). From Edward Snowden's revelations to the extraction practices of tech giants like Meta, X, and Google, these technologies have pushed us into an unprecedented era of pervasive technology-enabled surveillance—affecting citizens' lives not only in autocratic states, but also in democratic societies (Murakami Wood & Wright, 2015; French & Monahan, 2020; Dall'Agnola, 2024).

As digital technologies evolved—from basic ICTs to AI and ubiquitous CCTV—it seems that political rights and civil liberties are being eroded. Numerous indices point to the global resurgence of authoritarianism and the weakening of the democratic idea. Nearly 72 percent of the world's population now lives under autocratic rule, and for the first time in more than two decades, the world has fewer democracies than autocracies (V-DEM Democracy, 2025). Although technological tools were originally envisioned as empowering instruments that would promote democratic values and strengthen civil society across borders (Axford, 2011; Ghonim, 2012; Onuch, 2015), political and civil rights are declining in more countries than they are improving (Freedom of House Report, 2025). These developments stand in stark contrast to the optimism of the early post-Cold War period, epitomized by Francis Fukuyama's (1989) “end of history” thesis. Instead, the last two decades have ushered in a “democratic recession” (Diamond, 2015) and a technology-enabled

wave of re-authoritarianization, in which countries once considered democratic frontrunners are now “at risk” or have consolidated authoritarian rule (Brownlee, 2007; V-DEM Democracy, 2025).

This technology-enabled wave of re-authoritarianization has sparked a proliferation of terms—such as digital authoritarianism, techno-authoritarianism, networked authoritarianism, digital repression, and digital illiberalism—intent on capturing technology-enabled surveillance as a tool of authoritarian consolidation and democratic backsliding (MacKinnon, 2011; Glasius, 2018; Feldstein 2019, 2021; Polyakova & Meserole, 2019; Dall'Agnola, 2024). Yet, these concepts are often inconsistently defined and unevenly applied across regime types. While some scholars restrict their usage to non-democratic states (MacKinnon, 2011; Feldstein, 2019; Deibert, 2020), others emphasize that technology-enabled surveillance practices are also proliferating within hybrid and democratic regimes (Glasius, 2018; Akbari, 2022; Dall'Agnola, 2024). The result is a fragmented conceptual landscape that limits comparative insight and policy relevance.

This conceptual diffusion reflects a deep gap in our discipline: the absence of systematic theory-building and empirical study of technology-enabled surveillance (Raab, 2018; Akbari & Murakami Wood, 2025). As Raab (2018) observes, “there is little sustained effort within political science to analyse [technology-enabled] surveillance and related practices affecting privacy and other rights and freedoms against the background of political theories about power, democracy, fairness, and the liberal state” (p. 258). While adjacent disciplines such as law, sociology, science and technology studies (STS), and media studies have developed a rich body of literature on technology-enabled surveillance, political science has largely marginalized such inquiry. Recent scholarship suggests this neglect originates in the discipline's longstanding normative-democratic orientation, which obscures the infrastructural mechanisms of authority and control (Raab, 2018; Akbari & Murakami Wood, 2025). As a result, technology-enabled surveillance is often being considered insignificant to institutional politics or is being relegated to the subfield of security studies. This means that political science does not only seem to be unable to understand the attractiveness or pull force of authoritarian regimes, but also the ongoing decline of democratic institutions prompted by technology-enabled mechanisms of control.

In response to this blind spot, our introduction proposes a theoretical addition to Juan Linz and Alfred Stepan's (1996) seminal typology of regime types and types of path dependency. We propose adding a fifth regime characteristic: technology-enabled surveillance. We argue that while Linz and Stepan's framework remains analytically robust, it predates the Internet revolution and overlooks how surveillance practices enabled by ICT, AI, and CCTV technologies are shaping the interactions of states and societies. The rapid expansion of technology-enabled surveillance infrastructure has made monitoring, data extraction and algorithmic control structurally significant features of governance in both democratic and non-democratic regimes. This shift demands a rethinking of how regime types are conceptualized and compared.

To illustrate the empirical relevance of our conceptual extension, the contributions to our special section focus on a region that has experienced both rapid digitalization and re-authoritarianization in the last two decades: Central Asia (Dall'Agnola & Wood, 2022; Kassenova & Frigerio, 2024; V-DEM Democracy, 2025). The region offers both comparative variation of regime types and a unique opportunity to observe technology-enabled surveillance dynamics up close. As a result, the contributions to our special section explore surveillance not merely as a technological tool, but as an increasingly essential regime characteristic.

In the next section we briefly revisit Linz and Stepan's (1996) typology, before we conceptualize technology-enabled surveillance as a fifth regime characteristic and examine how its forms, functions, and influence vary across both democratic and non-democratic regimes. In the final section, we discuss the choice of Central Asia as a regional case and present the dominant themes that emerged from the contributions to this special section.

TECHNOLOGY-ENABLED SURVEILLANCE AS A KEY REGIME FEATURE IN THE DIGITAL ERA

Linz and Stepan's *Problems of Democratic Transition and Consolidation* (1996) offers one of the most influential frameworks for analyzing political regimes beyond the democratic-authoritarian binary. The authors identify four key dimensions that distinguish political regimes: pluralism, ideology, leadership, and mobilization.

The first characteristic, pluralism, refers to the degree of autonomy granted to political, social, and economic actors. In democracies, pluralism is institutionalized and constitutionally protected. Totalitarian regimes, by contrast, seek to eliminate it (Linz & Stepan, 1996, p. 156). The idea of establishing total control is an essential feature of totalitarianism, as Hannah Arendt argued in her seminal *The Origins of Totalitarianism* (1973). Under systems like Stalinism or Nazism, civil society was dismantled and individuals atomized (Baer, 2022). Post-totalitarian regimes inherit the institutional architecture of totalitarian control but begin to tolerate limited pluralism in practice—often for pragmatic reasons rather than intentions at reform. These concessions, however, remain tightly monitored (Linz & Stepan, 1996, pp. 156–158). In authoritarian regimes, pluralism is selectively restricted: political competition is constrained, but considerable autonomy may be allowed in the economic or social realms (Linz & Stepan, 1996, p. 156).

Ideology, the second characteristic, plays a central role in totalitarian regimes. According to Arendt (1973, p. 345, p. 371), totalitarian ideology pursued a distinctive goal: “The aim of totalitarian education has never been to install convictions but to destroy the capacity to form any” (p. 468). Ideology serves not merely as a legitimation strategy but as an all-encompassing system that explains the world in its entirety, or rather, totality (Baer, 2022). Linz and Stepan (1996, pp. 160–161) consider this ideological dimension as the core distinction between totalitarian and non-totalitarian regimes. In post-totalitarian regimes, ideology persists but loses its mobilizing power. Citizens participate in symbolic performances not out of belief, in the state ideology, but due to habit, coercion, or fear. Authoritarian regimes are less ideologically driven and often lean on nationalism, tradition, or performance legitimacy (Linz & Stepan, 1996, pp. 160–161).

The third dimension, leadership, concerns the nature of political authority and the degree of its institutionalization. Totalitarian leaders, such as Hitler, Stalin or Mao, are typically charismatic figures who embody the regime's ideological mission. As Arendt (1973) notes, the Leader has the “monopoly of knowledge” (p. 408)—a status that transforms deviation into treason against the supposed historical destiny of the nation. Post-totalitarian regimes, however, shift away from charismatic leadership toward bureaucratic rule, with party elites managing day-to-day governance (Linz & Stepan, 1996, pp. 159–160). Leadership in authoritarian regimes varies—from dominant-party systems to military juntas to family clans—but authority tends to be shaped by elite pacts or informal norms, while ideology is existing as a superficial aspect of legitimacy.

The final characteristic, mobilization, concerns the regime's efforts to engage citizens in political life. According to Linz and Stepan (1996) totalitarian regimes pursue extensive and compulsory mobilization, aiming to transform private individuals into political subjects aligned with the regime's goals (p. 161). In a totalitarian regime, citizens are forced to participate in rallies and engage in denunciation to affirm their loyalty to the state. Post-totalitarian regimes retain mass organizations and symbolic rituals, but participation becomes increasingly ritualistic and less emotionally charged, since the revolutionary fervour is gone (Linz & Stepan, 1996, p. 161). Citizens usually comply out of fear or habit, but the regime no longer expects genuine ideological commitment (Linz & Stepan, 1996, p. 161). Authoritarian regimes typically discourage political engagement altogether, preferring disinterest or political apathy. They do not want to convince citizens of a new dawn of mankind but demand obedience – that is enough for them to rule.

Linz and Stepan's typology remains a seminal, foundational tool for understanding regime change. However, it reflects the political context of the late twentieth century—a world where analogue forms of control dominated. What their framework does not explain – and how could

the authors possibly have predicted the development of technology? – is how the rapid expansion of ICT, AI, and CCTV technologies has made surveillance a structurally significant feature of governance in both democratic and non-democratic contexts. In what follows, we examine how technology-enabled surveillance operates within each of Linz and Stepan's regime types, illustrating both continuities and divergences in its purpose, scope, and institutional configuration.

In democracies, technology-enabled surveillance is often justified as a crucial tool for governance, particularly in the domains of public health, safety, national security, and law enforcement (Haggerty & Samatas, 2010; French & Monahan, 2020; Kitchin, 2020; Dall'Agnola, 2024). In times of crisis, such as terrorist attacks or pandemics, surveillance measures like CCTV surveillance or police wiretaps often scale up rapidly, with limited public debate (French & Monahan, 2020; Kitchin, 2020). These practices tend to be selective, targeting perceived threats such as terrorism and criminal activities, and their effects are unevenly distributed across the population (Monahan, 2022; Dall'Agnola, 2024). While technology-enabled surveillance in democracies is not without limits and scandals, such as Europe's PegasusGate (Mildebrath, 2022), they often prompt political backlash, public inquiries, and regulatory responses (see the EU's General Data Protection Regulation, GDPR). In parallel, technology-enabled surveillance in democracies increasingly extends beyond direct state control to private corporations (Zuboff, 2019, 2022; Akbari, 2022). Companies like Google, Meta, and telecom providers collect vast amounts of personal data, which is often used for political manipulation, as well as for commercial purposes (Zuboff, 2019, 2022; Akbari, 2022; Laruelle, 2023; Dall'Agnola, 2024). This data is largely collected outside democratic oversight and often repurposed for state surveillance and political interference (Haggerty & Samatas, 2010; Cuenca & Caro, 2025; Livingston & Miller, 2024). High-profile scandals, such as Snowden's revelations and the Cambridge Analytica affair, have exposed how corporate infrastructures can be repurposed for technology-enabled surveillance (Zuboff, 2019, 2022).

In non-democratic regimes, technology-enabled surveillance functions as a central tool of political control, designed to suppress dissent, stifle democratic organization, and reinforce regime authority (Morozov, 2011; Feldstein, 2021; Akbari, 2022). While methods vary, technology-enabled surveillance in these systems is both targeted and generalized: it monitors perceived threats—such as activists, journalists, or minority groups, as well as society as a whole.

Totalitarian regimes deploy extensive and centralized technological surveillance systems that penetrate all aspects of life. In a totalitarian regime like North Korea, surveillance is overtly ideological—designed to enforce absolute conformity through informant networks, forced self-criticism sessions, and relentless propaganda (Baek, 2024). In addition to these human-centered mechanisms, the state employs advanced technologies: smartphones issued by the regime are equipped with spyware, unauthorized files are auto-deleted without state signatures, and citizens are restricted to a closed intranet (*Gwangmyeong*) that isolates them from the global web (Baek, 2024).

Post-totalitarian regimes such as China retain many of these structures but adapt them with greater technological sophistication. China's 'Great Firewall' censors foreign content and controls internal discourse (MacKinnon, 2011; Lee, 2019; Deibert, 2020), while technology-enabled surveillance programs such as the Integrated Joint Operations Platform (IJOP) combine facial recognition, phone tracking, and biometric data to monitor individuals—particularly ethnic minorities like the Uyghurs—in real time (Liyanage, 2024). Unlike the more overt technology-enabled surveillance of totalitarian regimes, these systems are layered, automated, and partially invisible, embedding control into everyday social infrastructures. While less ideologically rigid than North Korea, China's technology-enabled surveillance regime is no less pervasive or repressive.

As our contributors to this special section show, the authoritarian regimes of Central Asia also make use of technology-enabled surveillance to pre-empt dissent and consolidate executive power, but with greater flexibility and less doctrinal ambition than China and North Korea. Media censorship, spyware, targeted intimidation and strategic internet shutdowns are used to

TABLE 1 Technology-enabled surveillance in different regime types.

Regime Type	Pluralism	Ideology	Mobilization	Leadership	Technology-enabled Surveillance
Democracy	Extensive	None or limited	Voluntary	Accountable, open	Selective, targeted, state – + corporate-led
Totalitarianism	Eliminated	Strong, guiding	Extensive, forced	Charismatic, central	All-encompassing, ideological + state-led
Post-Totalitarianism	Emerging	Ritualized	Declining	Bureaucratic	Very prominent, automated, partially invisible, state-led
Authoritarianism	Limited	Weak or absent	Low	Predictable, not static	Selective, strategic, opportunistic, state-led

disrupt protest coordination and suppress oppositional narratives (Anceschi, 2015; Dall'Agnola & Wood, 2022). In these contexts, governments use technology-enabled surveillance as an opportunistic instrument rather than as systematized means of oppression of the entire population.

Table 1 below summarizes how technology-enabled surveillance operates differently within each regime type, highlighting its varied scope, methods, and institutionalization across democratic, totalitarian, post-totalitarian, and authoritarian systems.

CENTRAL ASIA AS A REGIONAL CASE STUDY

To illustrate the empirical relevance of technology-enabled surveillance as a new regime characteristic, the contributions to this special section focus on Central Asia. Why? Because the region has undergone both rapid digitalization and re-authoritarianization in the last two decades (Dall'Agnola & Wood, 2022; Kassenova & Frigerio, 2024). While the five Central Asian states, Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, and Uzbekistan, initially adopted democratic frameworks after gaining independence from the Soviet Union—such as elections, multiparty systems, and parliaments—by 2025, they have all consolidated varying forms of non-democratic rule. These range from closed authoritarian regimes in Turkmenistan and Tajikistan to hybrid authoritarian regimes in Kazakhstan and Uzbekistan (Kassenova & Frigerio, 2024; V-DEM Democracy, 2025). Even Kyrgyzstan, once considered an “island of democracy” in the region, has seen a steady erosion of press freedoms, political opposition, and academic autonomy since the election of President Sadyr Zhaparov in 2020 (Panier, 2024; Daminov, 2025). As the contributions to this special section show, these regimes use various digital tools—from ICT, AI, to CCTV—to monitor dissent, control information, and extent autocratic power. Central Asia, therefore, presents a critical case for examining how technology-enabled surveillance shapes the nature, resilience, and global influence of non-democratic regimes in the 21st century.

Another key reason to focus on the Central Asian region is its relative accessibility to research on technology-enabled surveillance practices. Much of what we know about technology-enabled surveillance in non-democratic contexts remains speculative, based on limited reporting, state-controlled narratives (policy documents, official statistics, media reports), and reported insights from dissidents and defectors (Majerowicz & de Carvalho, 2024; Vila Seoane & Álvarez Velasco, 2024).

That being said, conducting research in Central Asia presents its own challenges, shaped by the region's authoritarian context in which scholars work (Dall'Agnola & Sharshenova, 2024). One of the primary challenges is navigating the tension between conducting rigorous academic research and protecting both the researcher and the

participants from the risks associated with state surveillance (Gentile, 2013). With systems such as state-sponsored CCTV networks, as well as advanced surveillance technologies like Pegasus and FinFisher spyware, researchers face an environment where surveillance is not only ubiquitous but also an obstacle to fieldwork (Dall'Agnola, 2023). As a result, researchers must be strategic, adapting their methods to minimize risks and avoid detection. Many scholars resort to self-censorship, limiting their online presence, reducing the scope of public discussions, and even withholding controversial research findings (Honig, 2024; Sharshenova, 2024). In some cases, researchers may even publish under pseudonyms or delay or even cancel publication entirely to protect themselves and their participants from retaliation (Bekmurzaev et al., 2018; Honig, 2024; Norov, 2024; Anonymous, 2025).¹ This self-censorship practice is particularly intense among local scholars, who must walk a fine line between doing their job in an open and honest manner, and keeping themselves and their families safe (Norov, 2024; Sharshenova, 2024). As a result, apart from one local contributor who still lives and works in Central Asia, all other contributors, even those originally from the region, are now based abroad. This highlights the tangible and intangible fears faced by Central Asian political scientists, particularly the risk of prosecution for writing critically in an increasingly tense political environment, a fear that forces many scholars into self-exile (Norov, 2024).

Our special section opens with a research note by Colin Knox and Bakhytzhon Kurmanov (2025), which offers an accessible entry point for readers less familiar with the region. Drawing on Feldstein's (2019) taxonomy of digital repression and empirical data from the Digital Society Survey (Mechkova et al., 2022), the authors provide a comparative overview of how surveillance capabilities vary across Central Asia. Although the Freedom of House report (2025) classifies all five Central Asian states as "consolidated authoritarian regimes," Knox and Kurmanov show that the digital surveillance repertoires deployed by autocrats differ markedly. While all regimes in the region employ some combination of online trolling, political content censorship, arrests for online dissent, and the dissemination of false information, the intensity and composition of these practices vary. Their findings reveal a digital repression gradient that runs from Turkmenistan, the most repressive, to Kyrgyzstan, the least. Notably, Kazakhstan, Uzbekistan, and Kyrgyzstan rely more on soft-monitoring tactics such as disinformation and social media filtering, while Turkmenistan and Tajikistan lean more heavily on hard-monitoring mechanisms such as malware and targeted repression.

Extending this comparative perspective, Ildar Daminov (2025) investigates the underlying conditions that make certain autocratic regimes more likely to adopt digital tools for covert repression. He asks: what are the necessary conditions that lead autocratic leaders to use digital technologies for covert repression online? To answer this question, he employs a Qualitative Comparative Analysis (QCA) of 83 non-democratic regimes, supplemented by in-depth case illustrations from Kazakhstan and Kyrgyzstan. His QCA finds that digital uptake alone is not sufficient for high levels of covert repression; instead, it is regimes with a prior history of overt repression that are most likely to weaponize digital tools for surveillance. Daminov's case studies of Kazakhstan and Kyrgyzstan reinforce the QCA's insight, in as much as they show that digital surveillance is most pervasive in Central Asia not where digital tools are most available, but where they build upon pre-existing structures/histories

¹For instance, originally we planned to publish a special issue rather than a special section. However, several researchers declined invitations for fear their contributions might attract unwanted attention from the authorities, potentially endangering both their participants and loved ones. One of the more senior scholars we reached out to replied that they were no longer comfortable writing about technology-enabled surveillance in their country of expertise because their last publication on the issue had triggered the anger of the authorities, who then went on to publish the school address of their children on the Internet. Fearing that the authorities could inflict harm on their children, they decided to change the focus of their work to a topic that was less unpopular with the regime.

of authoritarian control. In Kazakhstan, a long-standing authoritarian legacy has facilitated the emergence of a domestic market for social media surveillance, troll and bot armies to intimidate online users, and state-backed disinformation campaigns—despite the continued reliance on manual censorship. In contrast, Kyrgyzstan's historically weaker coercive state apparatus has limited the scope and sophistication of digital surveillance. While technology-enabled surveillance did intensify after 2020, there is still no significant domestic surveillance industry.

While Daminov focuses on the domestic preconditions for technology-enabled surveillance, Edward Lemon and Oleg Antonov (2025) shift attention to the international models that Central Asian autocrats seek to emulate. While China and, to a lesser extent, Russia are often presented as paradigms of non-democratic internet governance, little empirical research has explored how their policies and technologies diffuse into other non-democratic states. Lemon and Antonov (2025) address this gap by investigating Tajikistan's legal and technological alignments with these two powers. Using textual analysis combined with plagiarism detection software, they compare twelve internet-related laws from China, Russia, and Tajikistan. Their findings reveal a significant degree of legal convergence: Tajikistan has closely adopted Russian—and, to a lesser extent, Chinese—information security laws. Beyond legal borrowing, Lemon and Antonov (2025) show how Tajik authorities are replicating Russia's use of bots, trolls, and state-sponsored bloggers to suppress dissent online, and have implemented the Russian System of Operative-Search Measure (SORM)² system to wiretap and monitor internet communications. At the same time, Chinese influence is evident in the deployment of technological surveillance infrastructure, including telecommunications systems and smart city technologies.

Concluding the special section, Jasmin Dall'Agnola (2025) turns from policy and implementation to perception of technology-enabled surveillance, offering an in-depth analysis of how Chinese surveillance cameras are viewed by urban publics in Central Asia. While much has been written about the geopolitical implications of Chinese tech surveillance exports to other non-democratic regimes (Majerowicz & de Carvalho, 2024; Vila Seoane & Álvarez Velasco, 2024), little is known about how these technologies are viewed on the ground in Central Asia. Drawing on interviews and opinion polling data from across the region from the Central Asia Barometer, Dall'Agnola shows that attitudes toward Chinese surveillance infrastructure are shaped less by concerns about digital privacy than by two historically rooted and geopolitically conditioned perceptions of authority and progress. First, many Central Asians do not view technology-enabled state surveillance as a violation of privacy, but as a normalized feature of public life—shaped by Soviet-era legacies of collectivism and routine state monitoring. Second, as China becomes a key provider of infrastructure and technology, Chinese surveillance equipment is increasingly seen as a necessary part of modernization. Rather than contesting these systems, many Central Asians accept them as a pragmatic trade-off for economic and technological benefits. Together, these patterns suggest that support for Chinese surveillance technologies is shaped less by ideological alignment than by colonial habitus and growing geopolitical dependency.

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²A Soviet-era intelligence technology which was initially designed to monitor phone calls but later adapted for the digital age (Lefebvre & McDermott, 2008).

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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